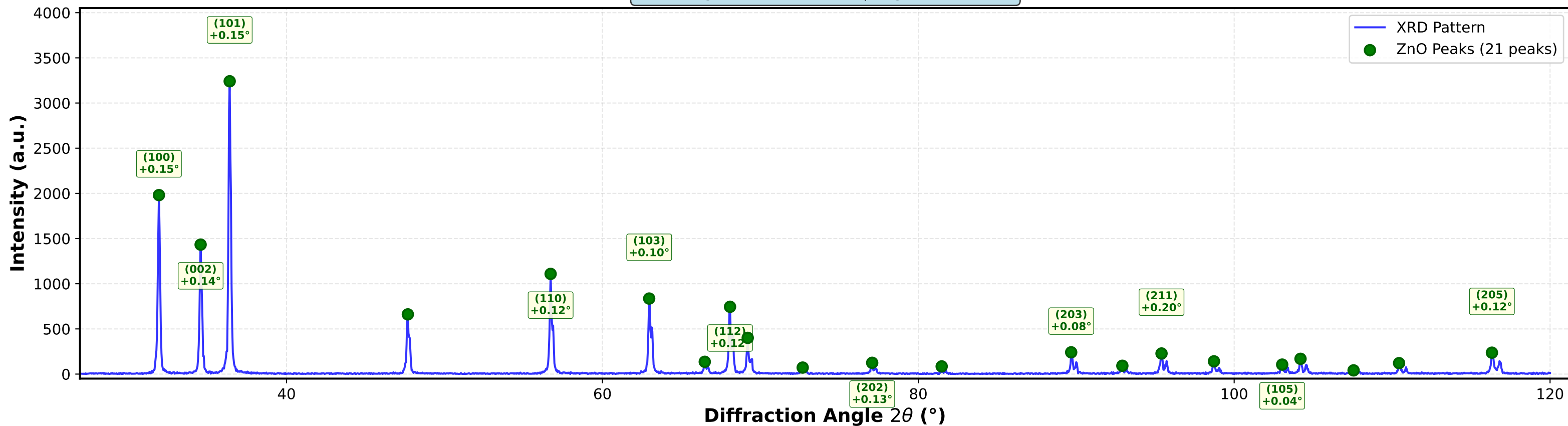


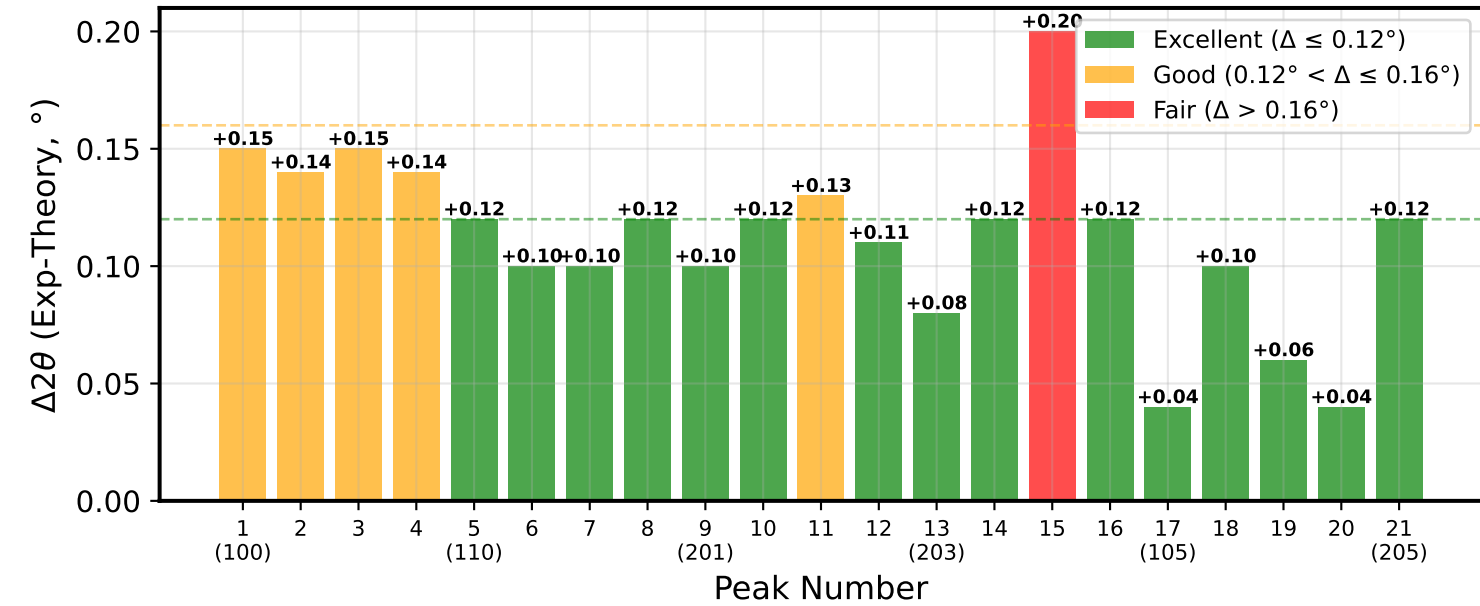
# Complete XRD Analysis Report: ZnO Wurtzite Structure

## XRD Pattern of ZnO Wurtzite Structure

2θ Range: 31.9° to 116.3° | Avg Δ2θ: 0.112°



### Peak Position Deviation Analysis



### Detailed Peak Information (Part 1/2)

| No. | 2θ_exp | 2θ_theo | Δ2θ  | hkl | Int. |
|-----|--------|---------|------|-----|------|
| 1   | 31.92  | 31.77   | 0.15 | 100 | 1981 |
| 2   | 34.56  | 34.42   | 0.14 | 002 | 1433 |
| 3   | 36.40  | 36.25   | 0.15 | 101 | 3241 |
| 4   | 47.68  | 47.54   | 0.14 | 102 | 661  |
| 5   | 56.72  | 56.60   | 0.12 | 110 | 1109 |
| 6   | 62.96  | 62.86   | 0.10 | 103 | 836  |
| 7   | 66.48  | 66.38   | 0.10 | 200 | 136  |
| 8   | 68.08  | 67.96   | 0.12 | 112 | 745  |
| 9   | 69.20  | 69.10   | 0.10 | 201 | 401  |
| 10  | 72.68  | 72.56   | 0.12 | 004 | 72   |

### Detailed Peak Information (Part 2/2)

| No. | 2θ_exp | 2θ_theo | Δ2θ  | hkl | Int. |
|-----|--------|---------|------|-----|------|
| 11  | 77.08  | 76.95   | 0.13 | 202 | 125  |
| 12  | 81.48  | 81.37   | 0.11 | 104 | 85   |
| 13  | 89.68  | 89.60   | 0.08 | 203 | 241  |
| 14  | 92.92  | 92.80   | 0.12 | 210 | 92   |
| 15  | 95.40  | 95.20   | 0.20 | 211 | 228  |
| 16  | 98.60  | 98.60   | 0.12 | 114 | 141  |
| 17  | 103.04 | 103.00  | 0.04 | 105 | 165  |
| 18  | 104.20 | 104.10  | 0.10 | 212 | 169  |
| 19  | 107.00 | 107.00  | 0.06 | 300 | 41   |
| 20  | 104.20 | 104.10  | 0.10 | 213 | 121  |
| 21  | 116.30 | 116.20  | 0.12 | 205 | 237  |

### Error Analysis

Peak with Best Match:  
Peak 20 (213):  
Δ2θ = 0.040°  
Error = 0.036%

Peak with Worst Match:  
Peak 1 (100):  
Δ2θ = 0.150°  
Error = 0.472%

### Crystallographic Information

Sample: ZnO (Wurtzite)  
Crystal System: Hexagonal  
Space Group: P6<sub>3</sub>mc  
Total Peaks: 21

Lattice Parameters:  
a = 2.8193 ± 0.0228 Å  
c = 5.1931 ± 0.0066 Å  
c/a ratio = 1.8419

Comparison with Standard:  
Δa/a = -13.25%  
Δc/c = -0.26%

Quality Assessment:  
✓ All peaks match ZnO structure  
✓ No impurity phases detected  
✓ Excellent crystal quality